

Intelligent Electricity Incident Reporting, Qualification & Monitoring Platform

Tripartite Collaboration Architecture: Citizens • Electricity Distributor (SOCADEL) • Authorized Contractors

Foundational Value Proposition

"Our application does not replace the electricity utility (SOCADEL). Instead, it acts as an operational intelligence layer that collects, aggregates, qualifies, and monitors incidents while strictly preserving SOCADEL's decision-making authority and operational workflow."

01 Operational Model & Intelligent Qualification

The system replaces raw, unorganized complaint queues with real-time spatiotemporal aggregation and clustering algorithms:

1. CITIZEN REPORTS (BASTOS SCENARIO)

18:42: 1st citizen reports a power outage
18:45: 15 additional reports within a 500m radius
Detection: Real-time spatiotemporal clustering
Action: Automatic clustering & synthesis

2. QUALIFIED INCIDENT CARD GENERATED

Incident: #2026-0047
Location: Bastos, Yaoundé (~2.4 km² estimated)
Reports: 37 correlated citizen reports
Severity: High | **Status:** Pending validation

02 SOCADEL's Central Role & Operational Authority

Unlike risky direct-dispatch models, the platform never automatically sends subcontractors to the field. SOCADEL retains full sovereign control to:

- **Validate or Reject** qualified incidents flagged by algorithm clustering.
- **Prioritize** critical urgency based on high/low voltage grid topologies and sensitive infrastructure.
- **Issue a Work Request** and assign authorized third-party contractors for field intervention.

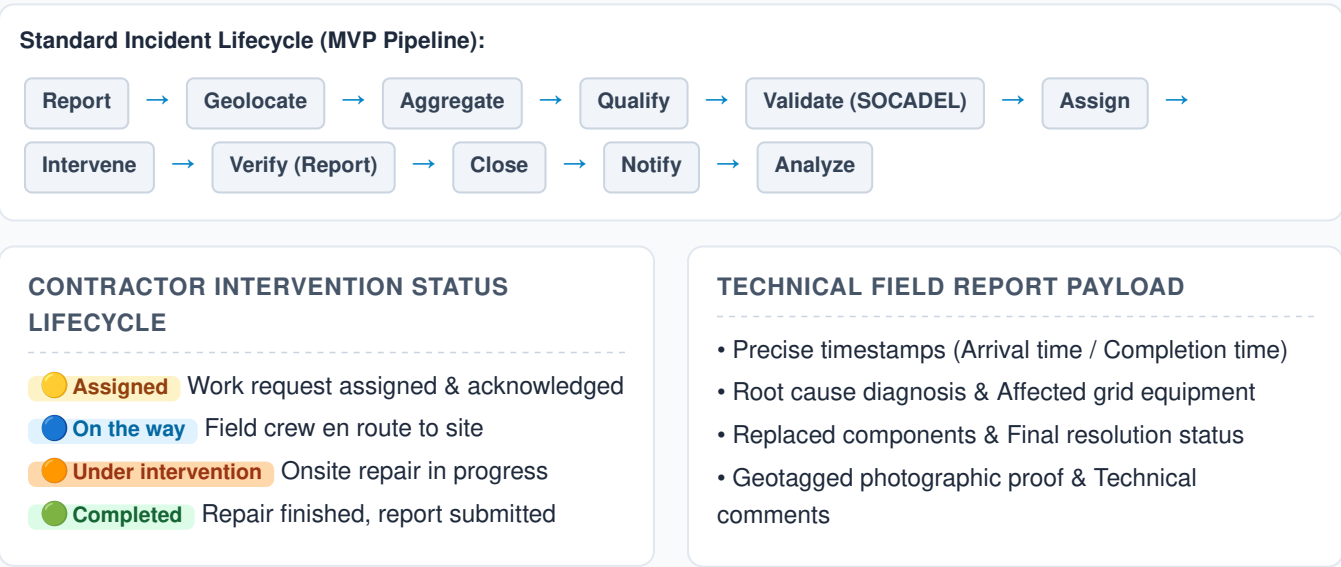
03 Actor Matrix & Clear Separation of Concerns

ACTOR	CORE ROLE	SYSTEM SCOPE & PERMISSIONS
 Citizen / Consumer	Reporter & Beneficiary	Reports outages (GPS, timestamp, category), tracks progress, and receives restoration notifications.
 SOCADEL Agent	Decision Authority & Dispatcher	Validates/rejects incidents, assigns priorities, generates Work Requests, and approves completion reports.
 Subcontractor	Field Execution Team	

ACTOR	CORE ROLE	SYSTEM SCOPE & PERMISSIONS
		Receives validated Work Requests, updates operational status in real-time, and submits field closure reports.
 System Administrator	Supervision & Governance	Manages user roles, configures spatial clustering thresholds, and oversees system performance/audits.
 Map / GPS Service	Geospatial Intelligence Provider	Provides reverse geocoding, computes outage polygon surface estimates, and powers live mapping interfaces.

04

End-to-End Operational Workflow



05

Academic Rigor & Defense Problem Statement

This architectural framing transforms the project from a generic utility app into an **enterprise-grade Decision-Support and Industrial Supervision Platform**.

RECOMMENDED CORE RESEARCH QUESTION (SOFTWARE ENGINEERING THESIS)

"How can an intelligent digital platform improve the reporting, qualification and monitoring of electricity incidents while preserving the authority and intervention workflow of the electricity distributor and its authorized contractors?"